

Sashank Erukala

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SUMMARY

Aerospace Engineer & Graduate Trainee with a strong foundation in UAV dynamics and control systems. **GATE 2025 Qualified (Rank 609)**. Proficient in Aerial Robotics development using **ROS 2, PX4, and Embedded Systems**, with hands-on experience in system integration, autonomous navigation and computer vision.

WORK EXPERIENCE

Junior Robotics Engineer — SkyTex Unmanned Aerial Solutions January 2026 – Present

- Integrated **APN guidance laws** into core PX4 modules for the Interceptor UAV flight stack.
- Developed **Striker**, a custom PX4 module to simplify integration of Interceptor UAV with external systems such as **ROS 2** and **MAVLink**.

Graduate Engineer Trainee — SkyTex Unmanned Aerial Solutions March 2025 – December 2025

- Used **Isaac ROS Visual SLAM and ORB-SLAM** running on edge hardware to obtain visual odometry and integrated it with **PX4** to enable reliable GPS-denied navigation.
- Analyzed PX4 flight logs to detect sensor anomalies, validating system performance prior to integration.

UAV Design Intern — NSIC June 2024 – July 2024

- Designed and fabricated agricultural UAV airframes, assembling heavy-lift propulsion systems for payload operations.
- Conducted stability flight tests to validate structural integrity and performance.

PROJECTS

Aerial Interceptor UAV & Custom GCS *ROS 2, C++, Qt*

Developed an interceptor algorithm for fixed-wing UAVs with APN guidance and PX4. Modified QGroundControl (QGC) source code to create a custom interface for real-time target interception.

GPS-Denied Autonomous Navigation *Isaac ROS, PX4, Embedded*

Integrated Isaac Ros Visual Slam with PX4 to fuse visual odometry for precision position hold and drift-free state estimation in GPS-denied environments.

AI-Powered Object Following Drone *YOLO, ROS 2, Python*

Developed a control loop with ROS2 and YOLO for realtime object detection and tracking by translating bounding box coordinates into velocity commands (Visual Servoing) for the PX4 flight stack.

ESP32 Flight Controller Design *KiCad, Embedded C++*

Designed and tested a custom 4-layer ESP32 flight controller. Developed the complete firmware stack including sensor fusion, stabilization and motor mixing.

PUBLICATIONS

2026 **Sizing of Multirotor Aerial Vehicle: A Statistical Approach** *Results in Engineering, Elsevier*

Developed a statistical sizing methodology for multirotor UAVs, correlating key design parameters across a wide dataset to enable rapid and accurate early-stage conceptual design.

2025 **Technical Framework for Human Detection Using Multirotor Aerial Vehicle in Virtual Environment** *ETA AV - 2025 Conference*

Proposed a simulation-based framework for detecting humans using UAVs, integrating PX4 autopilot with computer vision algorithms in a virtual environment.

CERTIFICATIONS AND ACHIEVEMENTS

2025	GATE 2025 Qualified	<i>All India Rank 609 (Aerospace)</i>
2025	Drone Systems and Control	<i>NPTEL (IISc Bangalore)</i>
2024	Deep Learning for Computer Vision	<i>NPTEL (IIT Hyderabad)</i>
2024	Introduction to Machine Learning	<i>NPTEL (IIT Madras)</i>
2019	NTSE Scholar (National Talent Search Examination)	<i>NCERT, Govt. of India</i>

EDUCATION

2021 - 2025	B.Tech Aerospace Engineering at Amrita School of Engineering	(CGPA: 8.27)
2019 - 2021	Class XI and XII at Sri Chaitanya Junior College	(Marks: 96.4%)
2018 - 2019	Class X at Sri Chaitanya High School	(GPA: 10.0)

SKILLS

Robotics	ROS 2, Isaac ROS, Isaac Sim, PX4, ArduPilot, Gazebo, NAV2, SLAM
Design	KiCad, Fusion 360, SolidWorks, MATLAB, Simulink, ANSYS
Programming	Python, C++, Embedded C, Qt, LaTeX